Chapter 11
Special Topics in Metrology



Metrology guides such as the JCGM 100 and JCGM 101 present in detail the most commonly used methods for evaluating measurement uncertainty. Additional guides for less commonly used methods are in development. In this chapter we present an overview of methods that we find valuable and deserving of more coverage. These methods include statistical process control (SPC), binary measurement systems (BMS), destructive measurements, sample size determination, and Bayesian analysis in metrology.

11.1 Introduction

This chapter discusses several special topics in metrology that have not received a great deal of emphasis in the metrology literature, but are considered important enough to cover here. The first special topic is statistical process control (SPC), in which the *process* of making a measurement is monitored using the tools of SPC. The second special topic is the evaluation of Binary Measurement Systems (BMS), in which the outcome of a measurement, such as with a go/no-go gauge, is binary. The third special topic is quantifying uncertainty when the measurement is destructive. The fourth special topic is sample size determination and allocation in metrology studies, and the final topic is Bayesian analysis in metrology.

11.2 Statistical Process Control (SPC)

Statistical process control (SPC) has traditionally been used in manufacturing to monitor and improve production processes. The goal of SPC is to quickly detect when a process upset has occurred and to eliminate the "assignable" cause. The same goal applies to the process of making measurements. Any significant departure of a

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the process standard deviation. Additional details for both of these charts are presented in Crowder (1987a) and Montgomery (2013).

Common control chart patterns (departures from nominal) can be detected with these two charts. The most common patterns include gradual changes in mean level, single unusual values, instability, sudden shifts in mean level, sustained trends up or down, and cyclical behavior. Each of these patterns may point to specific causes associated with the measurement process. It is in identifying causes of unusual behavior that subject matter expertise and statistical expertise come together.

In the case study in Sect. 11.2.1, an X − MR chart was used to monitor the performance of a battery tester used for product acceptance.

11.2.1 Case Study: Battery Tester Uncertainty and Monitoring Via SPC

In manufacturing applications, a check standard (Sect. 3.6.2) is often used to monitor tester performance over time. The check standard should be constructed as close as possible to the manufactured part being measured, and it should be a stable device that can replicate measurements over time. Measurements of the check standard are made on a frequent basis and can be plotted on an SPC control chart to verify ongoing tester performance. Any assignable causes of variation in the measurement process can then be detected via the control chart and eliminated to the extent possible.

The tester under investigation in this case study was a battery product acceptance tester that measured voltage output from four different channels of a battery. Because the testing of manufactured batteries is destructive, a check standard was used to simulate nominal voltages from the four channels. These voltages were measured by the tester at the rate of 10 readings per second for 15 s. The data retrieval was performed 50 times to establish a baseline for each of the four channels, resulting in 7500 ($10 \times 15 \times 50$) total measurements for each channel.

An $\overline{\text{X}} - \text{R}$ chart was first considered for monitoring tester performance. Prior to implementing this control chart for ongoing control, an uncertainty analysis of the tester was performed to establish baseline control limits. The initial control limits were based on the combined standard uncertainty of the battery tester determined from measurements of the check standard. Data from just the 10-volt channel were used in this case study.

The combined standard uncertainty of the battery tester was based on the Type A uncertainty of the tester combined with the Type B uncertainties of both the tester and the check standard. The standard deviation of the initial 7500 measurements of the 10-volt channel was $u_A(V) = 0.0141\ V$. Prior independent analysis of the check standard yielded $u_B(V) = 0.000171\ V$, while the Type B uncertainty of the tester was considered negligible. The combined standard uncertainty of the battery tester was thus

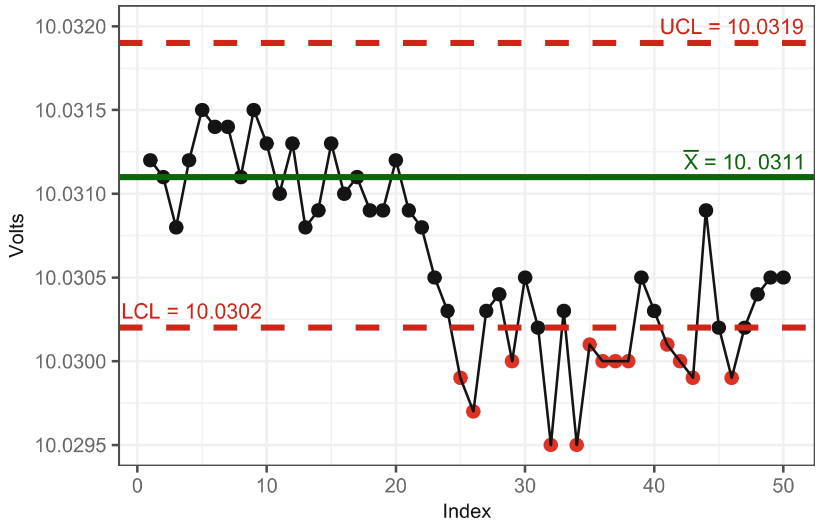


Fig. 11.3 Individuals control chart of ongoing voltage data

$$\widehat{\sigma} = \frac{\sum_{i=1}^{49} |x_{i+1} - x_i|/49}{1.128}.$$

The estimate $\widehat{\sigma}$ is based on the average of 49 moving ranges (from 50 observations) divided by the constant 1.128. This constant is the value that makes $\widehat{\sigma}$ an unbiased estimate of σ when the data are normally distributed (see Montgomery 2013). These baseline control limits were then used for an ongoing X − MR control chart of the subsequent 50 observations (Fig. 11.3). The control chart in Fig. 11.3 shows that initially the ongoing performance of the battery tester was consistent with baseline performance. However, around observation 20, the mean voltage quickly dropped to a new level, triggering "out-of-control" alarms denoted by the red observations. This shift was statistically significant, but was not considered practically significant, given the voltage specification limits (10 ± 0.1 V). Because the shift was not considered important, the battery tester continued to be monitored using the check standard and the technique described here without adjustment to the check standard or tester. If this shift in mean level was considered important, an investigation would be needed to identify the cause of the shift and necessary adjustments to the measurement process. The moving range chart, constructed for the baseline data, did not indicate any significant increase in tester variability. Construction of the moving range chart for the ongoing data is left as an exercise at the end of the chapter.

11.2.2 Discussion

The case study in Sect. 11.2.1 illustrates the use of control charts to monitor tester performance over time. Data from a well-designed, stable check standard were used

to establish baseline control limits for a battery tester. Ongoing data collected over time from the same tester measuring the same check standard were then monitored using a control chart with limits determined from the baseline data. The control chart used for ongoing control was an individuals chart with control limits based on the moving range ($X - MR$ chart). Other control charts could be used for this problem, including a cumulative sum chart (Lucas 1982) or an exponentially weighted moving average chart (Crowder 1987b, 1989). These alternative SPC control charts are more sensitive to subtle process changes than the $\overline{X} - R$ or $X - MR$ chart.

The control chart in this example was used to monitor the measurement process, not the manufacturing process. It should be noted that the measurement process can be stable and in-control even if the manufacturing process is out of control. These two processes can behave independently of each other, although the measurement uncertainty affects whether the manufacturing process is viewed as meeting specification limits (Chap. 5). If the measurement process remains in a state of statistical control over an extended period, the results could be used to assess tester drift and to establish calibration intervals for the tester (Chap. 10).

Many applications of SPC in metrology, such as the one above, involve the use of a check standard. When possible, it is preferable to measure manufactured parts over time, if the parts are stable and can be measured multiple times without degrading their output. The data collected over time can also be used to improve baseline estimates of repeatability and reproducibility (Sect. 9.3.4). In this case, the destructive nature of the test prevented repeated measures of actual batteries over time.

11.3 Binary Measurement Systems (BMS)

Binary Measurement Systems (BMS) are systems where the measurements can be classified into one of two categories. In engineering applications, these classifications are commonly 0/1, Pass/Fail, or go/no-go and are usually determined via a simple gauge or visual inspection. While BMS studies do not provide as much information as studies where the response is a continuous variable, they can be more efficient and are often used in practice Zhao et al. (2008). This section gives an overview of BMS studies and how to analyze the results, as well as guidance for choosing a sample size when designing a new study.

11.3.1 BMS Overview

For clarity, several aspects of a BMS study are defined below. A part is defined as the object being measured, an operator is the person taking the measurement, and a repetition is a repeated measurement of a single part. In a BMS study, several operators typically measure several parts, and each operator might measure each part multiple times. The total number of measurements, N, can be calculated as

$$N = o \times u \times m, \tag{11.3}$$

where o is the number of operators, u is the number of parts, and m is the number of repetitions of the measurement. For each inspection, the operator classifies the part using a binary outcome, such as "go" or "no-go."

There are two goals that are common in BMS studies: (1) assessing the agreement of operators by estimating the within-operator and between-operator agreement and (2) assessing the accuracy (correctness) of operators by estimating misclassification rates. Methods for addressing both goals will be discussed in the following sections. Guidance on choosing a sample size for a BMS study is provided in Sect. 11.3.4.

11.3.2 BMS Case Study Introduced

A case study is used to illustrate the concepts associated with evaluating a BMS. In the study, 3 operators measured 6 components at 20 different locations. For the purpose of illustration, each location on each component was considered an independent measurement, resulting in (effectively) 120 different parts. Each operator measured each part three times and provided a classification of Pass or Fail. Thus, the total number of inspections, N, was 360 for each operator. Table 11.2 provides a subset of the data for the component with serial number 2170.

11.3.3 Evaluation of a BMS

Assessing agreement and correctness is an important part of a BMS analysis as it provides a measure of the consistency of operators and the overall performance of the BMS. We will focus on the following three characteristics of a BMS:

- Within-operator agreement: the agreement within a single operator when repeated measurements are taken on the same part. Also known as intra-rater agreement.
- Between-operator agreement: the agreement between multiple operators when measurements are taken on the same part. Also known as inter-rater agreement.
- Correctness: the error rates (correctness rates) of an operator's classification of a part.

These metrics provide an indication of how the BMS is performing. If within-operator agreement or correctness is low for a given operator, that may be an indication that the operator requires additional training. Similarly, if the between-operator agreement or overall correctness is low, multiple operators may require additional training, or there may be issues with the gauge or the measurement process that need to be addressed.

Table 11.2 (continued)

Serial Number	Location	Rep	Operator 1	Operator 2	Operator 3
2170	15	1	Fail	Fail	Pass
2170	15	2	Pass	Pass	Pass
2170	15	3	Pass	Pass	Pass
2170	16	1	Pass	Pass	Pass
2170	16	2	Pass	Pass	Pass
2170	16	3	Pass	Pass	Pass
2170	17	1	Pass	Pass	Pass
2170	17	2	Pass	Pass	Pass
2170	17	3	Pass	Pass	Pass
2170	18	1	Pass	Pass	Pass
2170	18	2	Pass	Pass	Pass
2170	18	3	Pass	Pass	Pass
2170	19	1	Pass	Pass	Pass
2170	19	2	Pass	Pass	Pass
2170	19	3	Pass	Pass	Pass
2170	20	1	Fail	Fail	Fail
2170	20	2	Fail	Fail	Fail
2170	20	3	Fail	Fail	Fail

Table 11.3 Within-operator agreement for the BMS case study

Operator number	Number of parts inspected	Number in agreement	Proportion of agreement
Operator 1	120	113	94.2
Operator 2	120	116	96.7
Operator 3	120	114	95.0

11.3.3.1 Within-Operator Agreement

A straightforward method for assessing within-operator agreement is to tabulate the number of parts that agree (i.e., have the same classification for all repetitions) for each operator. This is referred to as the proportion of agreement within operators, and can be formally defined as

$$p_i = \frac{u_{ia}}{u}, \tag{11.4}$$

where u_{ia} is the number of parts that agree for the ith operator and u is the number of inspected parts. Table 11.3 provides the proportion of agreement for each operator in the BMS case study.

In this example, measurements agree if all three repetitions for a part were given the same classification (i.e., all Pass or all Fail) by an operator. Here, the operators have a high level of agreement across repetitions, implying that there is relatively

high precision for each operator. The acceptable level of within-operator agreement is problem specific, however, and may vary depending on the objectives of the analysis.

If more than three repetitions were taken, the requirement for agreement could be relaxed so that only a certain percentage (e.g., three out of four) of the classifications would need to be the same for the part to be in agreement, resulting in a less conservative estimate of the within-operator agreement.

11.3.3.2 Between-Operator Agreement

A basic method for assessing between-operator agreement is to determine how often the classification of different operators agrees across inspections. This proportion is defined as

$$p = \frac{n_a}{N}, \tag{11.5}$$

where n_a is the number of inspections that received the same classification from all operators and N is the total number of inspections. Table 11.4 shows the overall number of times all operators agreed on the classification of the part. Here, it appears that the operators are generally in agreement, with 345 of the inspections having the same classification across operators. While this method provides a high-level measure of between-operator agreement, it does not tell us if certain operators are more in agreement than others.

Cross-Tabulation The cross-tabulation method as described in Down et al. (2010) provides a more in-depth method for quantifying between-operator agreement. The first step is to assess the between-operator agreement for each pairwise combination of the operators. Table 11.5 shows the format of the cross-tabulation matrix that can be calculated for two operators. The first value in each cell, denoted n_{xy}, represents the number of times Operator A classified an inspection as x and Operator B classified an inspection as y. For example, n_{00} provides the number of times Operators A and B both classified an inspection as 0, a "Fail."

Table 11.4 Proportion of agreement for the BMS case study

Number of inspections	Number in agreement	Proportion of agreement
360	345	95.8

Table 11.5 Format of cross-tabulation matrix

		Operator B		
		Fail	Pass	Total
Operator A	Fail	n_{00} E_{00}	n_{01} E_{01}	n_0
	Pass	n_{10} E_{10}	n_{11} E_{11}	n_1
	Total	$n_{.0}$	$n_{.1}$	N

11.3.3.3 Assessing BMS Correctness

Often in a BMS study it is of interest to know the accuracy (correctness) of classifications. Correctness refers to whether an operator's classification of a part is equal to the "true state" of that part. At times, the "true state" of a part can be determined using a gold standard system (GSS). This section describes methods for assessing correctness when a GSS is available.

To formally define the methods for assessing correctness, let

$$X_{ijh} = \begin{cases} 1 \text{ if pass} \\ 0 \text{ if fail} \end{cases}$$

for $i = 1, \ldots, o$ operators, $j = 1, \ldots, u$ parts and $h = 1, \ldots, m$ repetitions. The "true state" can similarly be defined for the i*th* part as

$$Y_i = \begin{cases} 1 \text{ if conforming} \\ 0 \text{ if nonconforming} \end{cases}.$$

If a GSS is available, the "true state" is simply the classification provided by the GSS. It is assumed that the GSS value always classifies correctly. To assess BMS correctness, it is often of interest to estimate how often the system is incorrectly classifying parts, as well as what type of misclassifications are occurring. This can be accomplished by estimating the following two error rates:

$$\alpha = \Pr\left(X_{ijh} = 1 | Y_i = 0\right)$$

and

$$\beta = \Pr\left(X_{ijh} = 0 \middle| Y_i = 1\right).$$

Here, α is the probability the BMS passes a nonconforming part, while β is the probability the BMS rejects a conforming part. The misclassification rates α (also known as miss rate or false positive rate) and β (also known as false alarm or false negative rate) are often used to assess performance of a BMS. In many applications, having a low false positive rate is more important than a low false negative rate as it could be worse to pass a nonconforming part than to fail a conforming part. However, it is important to assess which rate will have a larger impact for a given application.

To estimate these rates, two roughly equal-sized samples (each of size N) of conforming and nonconforming parts (determined by the GSS) are taken and evaluated by having multiple operators classify the parts multiple times using the BMS under investigation. Each classification is then assigned one of these four labels:

Table 11.10 False positive and false negative rates for all operators in BMS case study

	α (False positive rate)	β (False negative rate)
Op. 1	0.022	0.003
Op. 2	0.006	0.008
Op. 3	0.019	0.003
Total	0.016	0.005

1. True positive—$X_{ijh} = 1$ and $Y_i = 1$.
2. True negative—$X_{ijh} = 0$ and $Y_i = 0$
3. False positive—$X_{ijh} = 1$ and $Y_i = 0$
4. False negative—$X_{ijh} = 0$ and $Y_i = 1$

The sample misclassification rates can then be calculated by summing the number of false positives (of N negatives) and false negatives (of N positives) over the total number of inspections in Eq. (11.10):

$$\alpha = \frac{\sum \text{False Positives}}{N}$$
$$\beta = \frac{\sum \text{False Negatives}}{N}. \tag{11.10}$$

Table 11.10 provides the false positive and false negative rates for the three operators, as well as an overall value across the operators. Operator 2 has a higher false negative rate, while Operators 1 and 3 have higher false positive rates. Overall, the operators have a higher false positive than false negative rate.

11.3.4 Sample Sizes for a BMS Study

This section addresses choice of sample sizes when designing a BMS study. With any estimate of agreement or correctness there is uncertainty due to the limited sample size used to generate the estimate. Consequently, a large enough sample size is needed to adequately assess a BMS given the constraints (time, money, etc.) that may limit the number of samples.

To assess how many classifications are needed, an analyst can quantify the uncertainty in the estimate of agreement and correctness for varying sample sizes. This section will describe how to calculate a confidence interval for the proportion of agreement p_0 between two operators. A similar analysis can also be done for a kappa value confidence interval (Blackman and Koval 2000) or a false positive rate confidence interval using the Binomial distribution.

A $100 \times (1 - \alpha)\%$ confidence interval for the proportion of agreement describes the uncertainty due to a finite sample size and can be calculated as

agreement moves closer to 50%, many more samples are needed to reduce the estimate's uncertainty. Based on the results in Fig. 11.4 and Table 11.11, the total number of inspections, N, should be at least 400 to ensure that the confidence interval width is less than 0.10 for all proportions of agreement. This sample size will be large enough to bound the confidence interval width of the correctness measure (described in Sect. 11.3.3.3) by 0.10, as well. To achieve this level of uncertainty, the number of operators, o, parts, u, and repetitions, m, should thus be chosen so that the product $o \times u \times m$ is at least 400. This number could be achieved, for example, using 50–60 parts with 2–3 operators and 4–5 repetitions of each part.

11.4 Measurement System Analysis with Destructive Testing

Many manufactured parts have properties that can be measured only once due to the destructive nature of the test. Examples include parts for which break strength, impact strength, or burst pressure may be measured. Examples within the NSE include neutron sources, detonators, and thermal batteries, in which both inputs that activate the device and outputs from the device can only be measured once. In cases like these, assessing the repeatability and reproducibility of the measurement becomes more difficult because the part is not available for repeated measurements. We have used two different approaches to this measurement problem, each with important assumptions.

The first approach uses an alternative source to produce the same type of output produced by an actual part. This approach was used in both the battery tester case study (Sect. 11.2.1) and the lead probe case study (Sect. 6.4.1). In the lead probe uncertainty study, an alternative neutron source was used to assess measurement uncertainty. In Fig. 11.5, the alternative source (with the yellow sticker) produces a neutron stream that is measured by a lead probe standard, located to the right of the source. Repeated shots of the source produced the data that were used in the lead

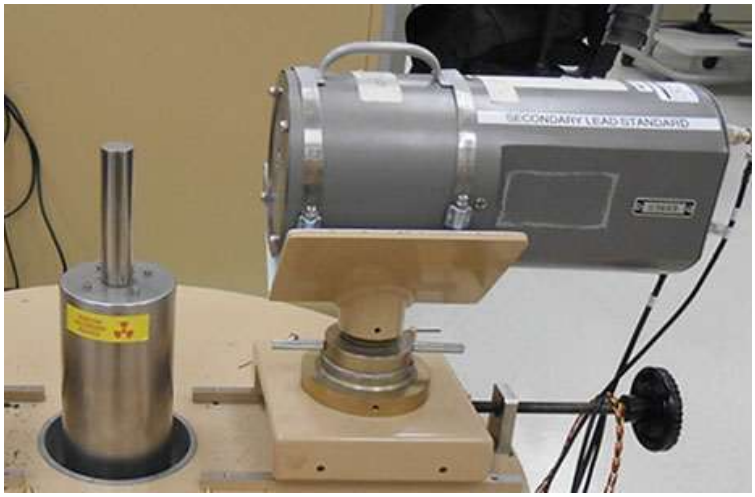


Fig. 11.5 Alternative neutron source under test

probe uncertainty analysis. The key assumptions with this approach are that the alternative source is repeatable and that its output adequately mimics that of an actual test unit.

The second approach to the destructive test problem relies on samples of manufactured parts being relatively homogeneous. Although not an ideal approach, it may be the best option available. Each part is measured, and resulting values are analyzed with the understanding that variation in the data includes part-to-part manufacturing variation. The resulting uncertainty analysis will be conservative in that part-to-part variation will inflate the estimate of measurement uncertainty. The same Gauge R&R methodology described earlier (Sect. 9.3.4) can be used, with repeated measurements of the same parts replaced by single measurements of different (homogeneous) parts. The key assumption with this approach is that the test units are homogenous, or that any significant non-homogeneity can be modelled and removed from the analysis. An example of this approach is given in Benham (2002). De Mast and Trip (2005) discuss a variety of approaches and important assumptions associated with Gauge R&R studies in which measurements are destructive. Both of these papers include a handful of additional references taken from a relatively small literature on this topic.

11.5 Sample Size and Allocation of Samples in Metrology Experiments

The expression for expanded uncertainty (Eq. 6.7)

$$U = t_p(\upsilon_{\text{eff}})u_c(y) \tag{11.12}$$

consists of two terms that are both greatly affected by the choice of sample size and sample allocation in a measurement uncertainty study. The first term is the multiplier $t_p(\upsilon_{\text{eff}})$, the $p \times 100$th percentile of the t-distribution with effective degrees of freedom υ_{eff}, often computed using the W-S approximation. The second term is the estimated combined standard uncertainty that is typically a combination of several Type A and Type B variance terms.

The GUM provides little guidance regarding how to choose an appropriate sample size and how to allocate those samples in a measurement experiment. This section presents simple approaches to determining the effect of sample size and sample allocation on the value of $t_p(\upsilon_{\text{eff}})$ and the precision in $u_c(y)$. The results provide objective criteria for assessing the relative value of a given sample size and allocation. These criteria can be used to determine a desirable sample size for most of the measurement experiments described in this book.

The impact of the degrees of freedom υ_{eff} on the multiplier $t_p(\upsilon_{\text{eff}})$, and hence on the expanded uncertainty, is easy to demonstrate graphically. For a 95% confidence interval, the value of $t_{95}(\upsilon_{\text{eff}})$ is shown as a function of υ_{eff} in Fig. 11.6.

judgement, we believe the relative uncertainty will rarely be greater than 20% (0.20), resulting in υ_B no less than 12.5. Using this conservative lower bound for the Type B degrees of freedom will not by itself decrease the overall υ_{eff} to an unacceptable level if the experiment has been properly designed.

The second approach discussed in the GUM (G.4.3) is to set the Type B degrees of freedom equal to $+\infty$. The GUM justifies this assumption by claiming it is equivalent to choosing a conservatively wide interval for the Type B uncertainty. If $+\infty$ is used, however, the effective degrees of freedom υ_{eff} will be artificially inflated whenever the Type B uncertainty term dominates the combined standard uncertainty. As an example, suppose $u_A(y) = 1$ with degrees of freedom υ_A and $u_B(y) = 2$ with degrees of freedom $\upsilon_B = +\infty$. Applying the W-S approximation,

$$\begin{aligned}\upsilon_{\text{eff}} &= \frac{\left(1^2 + 2^2\right)^2}{\left(\frac{1^4}{\upsilon_A} + \frac{2^4}{+\infty}\right)} \\ &= \frac{(5)^2}{\left(\frac{1}{\upsilon_A}\right)} \\ &= 25\upsilon_A. \end{aligned} \tag{11.22}$$

Setting the Type B degrees of freedom to $+\infty$ in this simple example significantly increases the effective degrees of freedom, exaggerating the actual amount of information obtained from the measurement experiment.

Hall and Willink (2001) have written in greater detail regarding the performance of the W-S approximation in cases like this. They show that the W-S approximation does not perform well whenever $u_B(y) \geq 2u_A(y)$ and the Type B term is assigned infinite degrees of freedom. When a Type B term dominates the combined standard uncertainty, the relative uncertainty approach (Eq. 11.21) should be used to assign Type B degrees of freedom. Assigning degrees of freedom $+\infty$ should only be done with caution, taking into account the impact on the total effective degrees of freedom.

11.6 Summary of Sample Size Recommendations

Based on the demonstrated effect of sample size on the value of $t_p(\upsilon_{eff})$ and on the precision of the estimate $u_c(y)$, a reasonable goal for a measurement uncertainty study is to achieve Type A effective degrees of freedom ($\upsilon_{A,\,eff}$) in the 20 to 30 range, with $\upsilon_{A,\,eff} = 15$ as minimally acceptable. Allocating the number of tests to each variable in the measurement equation based on the value of the individual uncertainties (the $u_i(y)'s$) can be used to optimize $\upsilon_{A,\,eff}$. Properly assigning degrees of freedom to each Type B term via Eq. (11.21) will result in acceptable yet realistic values for both $\upsilon_{B,\,eff}$ and the overall υ_{eff}. This is assured by the relationship

$$\min\left(v_{A,eff}, v_{B,eff}\right) \leq v_{eff} \leq \left(v_{A,eff} + v_{B,eff}\right),$$

where $v_{A,\,eff}$ and $v_{B,\,eff}$ are the effective degrees of freedom obtained from Type A and Type B uncertainty evaluations, respectively. These guidelines can be applied to sample size selection for basic uncertainty analyses (Chap. 6), test plans for factorial designs and ANOVAs (Chap. 9), and test plans for curve fitting (Chap. 10). Figures 11.6, 11.7, and 11.8 can also be used to investigate the increase in uncertainty associated with much smaller sample sizes than recommended here.

11.7 Bayesian Analysis in Metrology

The purpose of this section is to introduce the Bayesian approach to uncertainty analysis and to provide references for further study.

Since the publication of the GUM in 1995, Bayesian analysis has appeared regularly in the literature as an alternative to the classical frequentist uncertainty analysis emphasized in the GUM. The goal of a Bayesian uncertainty analysis is to incorporate prior belief regarding the unknown quantities of interest into the analysis. Experimental data is used to update the prior belief distribution to form a post-experiment, or posterior belief distribution about the unknown quantities. While the frequentist approach views the measurand as an unknown constant that is estimated from experimental data, the Bayesian approach views the measurand as a random variable whose distribution is derived from both prior knowledge and experimental data.

Prior knowledge is ideally available in the form of what is called an *informative* prior distribution. An informative prior can be developed from expert opinion, previous experimentation, or even from a previously derived posterior distribution. It is a distribution that reflects available knowledge prior to the collection of new data. When little or no prior information is available, Bayesian analysis may still be performed using what is called a *noninformative* prior. Noninformative priors are often taken to be simple constant values on the entire real line, although there is no universally agreed upon way to choose such a prior (Berger 1985, 2006). These priors are *improper* priors in that they cannot be normalized to integrate to one, a characteristic of proper probability density functions. This type of prior provides no information regarding the measurand or inputs to a measurement equation. A noninformative prior is used in the case of no prior information to preserve the mathematical framework of the Bayesian approach. Although the prior is improper, the resulting posterior distribution is proper.

A simple example of Bayesian inference for a normal distribution with known variance illustrates how the prior, sampling distribution (likelihood), and posterior distributions are related. Assume that a single measurement (x) is normally distributed with sampling distribution $f(x|\mu, \sigma^2)$, and that the prior distribution for the measurand μ is also normal with PDF $p\left(\mu|\mu_0, \sigma_0^2\right)$. Because the sampling distribution

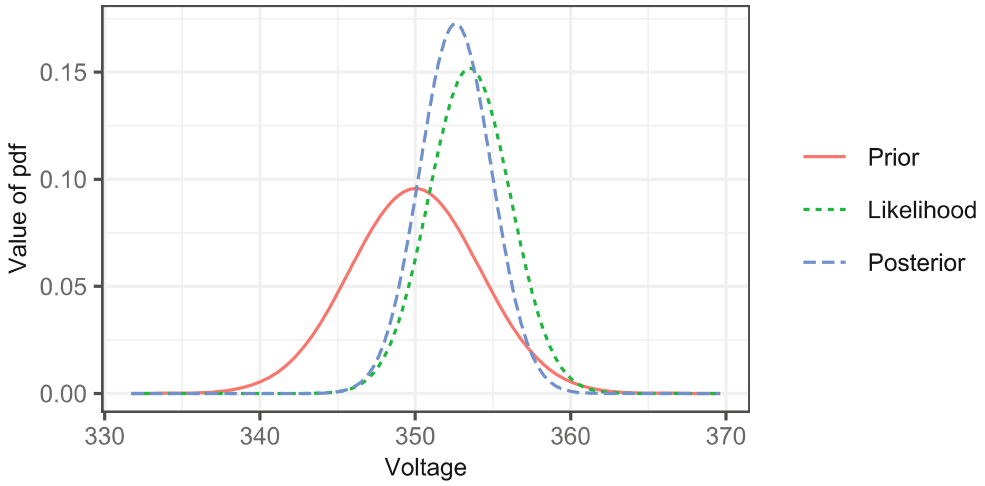


Fig. 11.9 Comparison of prior, likelihood, and posterior distributions

distribution will be a "weighted average" of the prior distribution, based on prior belief, and the sampling distribution, based on the measurement data. The weights given to the two distributions will depend on both the precision of the prior belief regarding the measurand, and the precision of the measurement data.

Lira and Grientschig (2010) present an assessment of Bayesian analysis in metrology and include several tutorial examples. The prior distribution is explained as a "state-of-knowledge" distribution carrying prior information about any supplementary and any input variables, *before* any data have been collected. The information provided by the measurement data is expressed in the form of a sampling distribution, and the resultant posterior distribution is expressed as proportional to the product of these two distributions. For measurement equations with even a few input quantities, the posterior distribution must be determined by a combination of mathematical analysis and numerical integration, but the availability of numerical analysis software makes this approach feasible. Several realistic metrology examples are presented, and uncertainty analyses are performed using this approach. This paper also provides numerous references from the literature that demonstrate Bayesian analysis in metrology.

11.8 Related Reading

Statistical process control (SPC) is not formally covered in the JCGM's guiding documents although the JCGM 100 (GUM, p. 7) states that "the use of check standards and control charts that can indicate if a measurement is under statistical control should be part of the effort to obtain reliable evaluations of uncertainty."

Guidelines for applying SPC to measurement problems appear in a variety of discipline-specific standards and regulations. These documents include ASTM D6299 (analytical measurements), ASTM E882 (chemical analysis), and ASTM

Case studies applying Bayesian analysis to metrology problems appear in the National Institute of Standards (NIST) research library (https://www.nist.gov/nist-research-library) and in the metrology journals *Metrologia, Measurement Science and Technology,* and *Measurement*. The JCGM Working Group 1 News Note of May 4, 2019, states that JCGM 108, to be developed at a later time, will cover "Bayesian Methods." No time frame was given for its development and release. While Bayesian analysis has not been presented in this book in detail, the authors recognize it as a promising methodology to use in measurement studies when important prior knowledge is available. We believe that as the methodologies are formalized for applications in metrology, and presented in JCGM guidelines, Bayesian analysis will move into the mainstream as a tool for practitioners of metrology.

11.9 Exercises

1. Give examples of both natural ("common cause") and unnatural ("assignable cause") variations present in a measurement process.
2. Construct a moving range (MR) chart for the "Ongoing Data" in Table 11.1. What do these MR charts say about the measurement process stability?
3. What is the difference between "statistical" significance and "practical" significance when interpreting an out-of-control condition on a control chart?
4. What is a binary measurement system? When might one be used?
5. What is the difference between within-operator agreement and between-operator agreement?
6. Use the cross-tabulation matrix provided in Table 11.15 to calculate the expected counts in each cell. Then calculate the between-operator proportion of agreement and Cohen's kappa value. Why might the Cohen's kappa value be advantageous over the between-operator proportion of agreement?
7. What is the difference between the false positive rate and the false negative rate? Which rate is worse?
8. Table 11.16 gives the results from a study that was performed to assess agreement and correctness in a BMS. In this study, three operators measured 20 parts each three times and provided a classification of 1 (Pass) or 0 (Fail). A GSS was available and used for all 20 parts.
   Calculate the proportion of agreement within operators. What does this proportion suggest? Next, calculate the proportion of agreement between operators. How does this compare to the within-operator proportion of agreement?

Table 11.15 Cross-tabulation matrix for two operators

	Op. 2: Fail	Op. 2: Pass	Total
Op. 1: Fail	40	4	44
Op. 1: Pass	12	120	132
Total	52	124	176

12. Assume that measurements of a counting process, X_1, X_2, . . ., X_n, are independent and identically distributed (i.i.d.) *Poisson* (λ) (Eq. 4.10). What is the likelihood function $L(\lambda|\, x_1, x_2, \ldots, x_n)$ for λ, given these data? If a *Gamma*(α, β) distribution is chosen as the prior for λ, what is the resulting posterior distribution for $\pi(\lambda|\, x_1, x_2, \ldots, x_n)$?

    (Note: The Gamma prior distribution for λ has the form: $p(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}$.)

    How would the posterior distribution for λ be used to construct a 95% uncertainty interval?

References

ASME Y14.43: Dimensioning and Tolerancing Principles for Gages and Fixtures. (2011)

ASTM D6299: Standard Practice for Applying Statistical Quality Assurance and Control Charting Techniques to Evaluate Analytical Measurement System Performance. (2013)

ASTM E882: Standard Guide for Accountability and Quality Control in the Chemical Analysis Laboratory. (2016)

ASTM MNL7: Manual on Presentation of Data and Control Chart Analysis. (2018)

Benham, D.: Non-Replicable Gauge R&R Study. Automotive Industry Action Group (2002)

Berger, J.: Statistical Decision Theory and Bayesian Analysis. Springer Series in Statistics. Springer, New York (1985)

Berger, J.: The case for objective Bayesian analysis. Bayesian Anal. **1**, 385–402 (2006)

Blackman, N.J.-M., Koval, J.J.: Interval estimation for Cohen's kappa as a measure of agreement. Stat. Med. **19**, 723–741 (2000)

Boyles, R.A.: Gauge capability for pass-fail inspection. Technometrics. **43**(2), 223–229 (2001)

Cohen, J.: A coefficient of agreement for nominal scales. Educ. Psychol. Meas. **20**, 37–46 (1960)

Crowder, S.V.: Computation of ARL for combined individual measurement and moving range charts. J. Qual. Technol. **19**(2), 98–102 (1987a)

Crowder, S.V.: A simple method for studying run length distributions of exponentially weighted moving average charts. Technometrics. **29**(4), 401–407 (1987b)

Crowder, S.V.: Design of exponentially weighted moving average schemes. J. Qual. Technol. **21** (2), 155–162 (1989)

Crowder, S.: An introduction to the Bernoulli CUSUM. In: Proceedings of the IEEE Annual Reliability and Maintainability Symposium, Orlando (2017)

Danila, O., Steiner, S.H., McKay, R.J.: Assessing a binary measurement system. J. Qual. Technol. **40**, 310–318 (2008)

Danila, O., Steiner, S.H., McKay, R.J.: Assessment of a binary measurement system in current use. J. Qual. Technol. **42**, 152–164 (2010)

Danila, O., Steiner, S.H., McKay, R.J.: Assessing a binary measurement system with varying misclassification rates using a latent class random effects model. J. Qual. Technol. **44**, 179–191 (2012)

Danila, O., Steiner, S.H., McKay, R.J.: Assessing a binary measurement system with varying misclassification rates when a gold standard is available. Technometrics. **55**, 335–345 (2013)

De Groot, M.: Optimal Statistical Decisions. McGraw-Hill, New York (1970)

De Mast, J., Trip, A.: Gauge R&R studies for destructive measurement. J. Qual. Technol. **37**(1), 40–49 (2005)

De Vor, R., Chang, T., Sutherland, J.: Statistical Quality Design and Control: Contemporary Concepts and Methods, 2nd edn. Pearson/Prentice Hall, Upper Saddle River (2007)

Down, M., Czubak, F., Gruska, G., Stahley, S., Benham, D.: Measurement Systems Analysis Reference Manual. Automotive Industry Action Group, Detroit (2010)